

Highlights

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

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- Transport adjoint importance and the rare-event committor are identical; verified to machine precision.
- Weighted Ensemble validated against exact enumeration; two decades deeper than naive Monte Carlo.
- FW-CADIS attains equivalent threshold coverage at roughly one-eighth the allocation cost.
- Learned importance coordinate gives a $1.19\times$ FOM ratio, bounded by a two-level bootstrap.
- Deep ensembling resolves that ambiguity as a decisive $0.087\times$ loss (95% CI [0.051, 0.155]).
- Naive Monte Carlo wins at every lattice size up to $L = 32$, with no narrowing margin.

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

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ABSTRACT

This work transfers the adjoint-importance toolkit of radiation shielding to spin systems, testing it end to end on the 2D Ising ferromagnet and the Edwards–Anderson spin glass. The identity between the transport adjoint and the rare-event committor is exact and is verified to machine precision. Weighted Ensemble, the statistical-mechanics analogue of split/roulette weight windows, is validated against exact enumeration and reaches two decades deeper into the tail than naive Monte Carlo at matched cost. A learned importance coordinate proves sensitive to network initialization: across seventeen independent retrains, a two-level bootstrap gives a geometric-mean figure-of-merit ratio of $1.19\times$ (95% CI [0.84, 1.75]) against the best hand-picked coordinate, a positive but not significant effect. A deep-ensemble follow-up, which attacks the diagnosed noise source (weight initialization) directly rather than adding more single-net seeds, reduces the run-to-run spread as intended but converges toward a loss; correcting a cost-accounting gap the ensemble exposes — neural-network inference cost had been silently uncounted — makes that loss statistically decisive ($0.087\times$, 95% CI [0.051, 0.155]), the only comparison in this paper whose confidence interval excludes parity. A label-free self-consistency objective from recent neural importance-function work likewise yields a favourable point estimate that does not survive bootstrap replication. FW-CADIS global allocation matches the error profile of uniform allocation at roughly one-eighth the simulation cost, although the flattening of relative error for which it is named is not observed at the scales tested. Naive Monte Carlo remains the most efficient estimator up to $L = 32$, placing the tested rarity regime below the crossover at which variance-reduction overhead is repaid. Together these results establish a validated transfer of the transport sampling machinery to spin systems, delimit its present practical limits, and supply a reproducible benchmark and statistical discipline for future work.

1. Introduction

Rare-event sampling is the shared computational difficulty of two literatures that rarely cite one another. Weighted Ensemble, Forward Flux Sampling and committor-based methods are already the working tools for nucleation, protein folding and reaction-rate calculation [1–3], and the obstacle they all face is structural: an unbiased estimator of an event of probability p has relative error of order $(Np)^{-1/2}$, so each decade of rarity costs an order of magnitude in computing [1, 4]. Every serious remedy therefore replaces more samples with better-placed ones, an idea present in Monte Carlo since importance sampling and trajectory splitting [5, 6].

Radiation transport met the problem first, in deep-penetration shielding, where analog scoring probabilities fall below 10^{-10} [7, 8]. Its remedy is the *adjoint* flux, or importance function, which runs the transport equation backwards from the detector response to give the expected contribution of a particle at each phase-space point [9, 10], and is then used both to bias source emission and to set weight windows, inside which particles are split and outside which they are killed by Russian roulette [5, 11]. CADIS automated this by deriving both devices consistently from a coarse deterministic adjoint, giving figure-of-merit gains of several orders of magnitude on realistic geometries [8, 12]; its forward-weighted extension FW-CADIS replaces the single-detector objective with a *global* one, seeking uniform uncertainty across a whole family of responses [13, 14]. Performance is judged throughout by the figure of merit $FOM = 1/(\sigma_{\text{rel}}^2 T)$, a cost-normalised inverse variance [7, 13].

Statistical mechanics reached closely related tools independently. Transition Path Theory describes reactive trajectories through the committor $h(s)$, the probability of reaching a target set before returning to the origin set, which is the optimal reaction coordinate in a precise sense [3, 15, 16]; its practical samplers are splitting methods in disguise, whether by splitting and merging weighted replicas across bins [17, 18], advancing an ensemble through fixed or adaptive interfaces [1, 19], or working directly in trajectory space [20]. What unites and limits them is dependence

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on a user-supplied coordinate, with no general prescription for choosing it [1, 2]. Lattice spins are the natural testbed for that limitation: the 2D Ising ferromagnet [21] has an exact solution to validate against [22], while the Edwards–Anderson model adds quenched disorder, frustration and no obvious coordinate [23, 24]. Both are usually simulated with Metropolis dynamics [25], whose critical slowing down motivated cluster updates [26, 27] and flat-histogram schemes [28, 29] — methods that flatten a free-energy landscape rather than steer trajectories toward a specified tally, and so optimise a mismatched objective here [29].

That the two importance functions coincide is classical rather than new: both are instances of Doob’s h -transform [30], which reappears as the driven process of large-deviation theory underpinning the zero-variance sampler [4, 31] and, in stochastic-control form, as the solution of a Gibbs variational principle [32]. Machine learning has supplied a third route. Physics-informed networks solve high-dimensional PDEs without a grid [33], and were applied to the committor equation itself [34, 35], with active importance sampling concentrating training effort where rarely visited regions dominate the loss [36]; a separate line trains a committor variationally and uses it as a metadynamics-like collective variable with a bias potential rather than a weight-window schedule [37]. Generative models were meanwhile trained directly on spin systems — variational autoregressive networks [38] and Boltzmann generators [39] — with later work recovering unbiased observables by reweighting [40] or embedding the model in a Markov chain [41].

The gap this leaves is specific. The identity is established [30, 31] and neural committors exist [34, 35, 37], but the transport *toolkit* built around that identity — consistent weight windows, the global FW-CADIS objective, the FOM as an evaluation standard — was designed for precisely the regime where learned-committor methods are weakest [8, 13], and has never been transferred to spin systems and audited. Every neural-committor line above samples by continuously reweighting or biasing dynamics rather than by the transport toolkit’s discrete population control: splitting a walker, killing it by roulette, resampling to a fixed per-bin occupancy. The closest prior work makes the point concretely. Kim and Cai [42] learn an importance function $I(s)$, modify transitions as $K'(i \rightarrow j) = K(i \rightarrow j)I(j)/I(i)$, and use a branching random walk — in the language of §2.1.3, their I is our h , their K' is the tilt, and their branching is splitting, so they independently reconstruct the zero-variance/weight-window recipe for discrete rare transitions without the transport framing, and without comparing their learned coordinate against a hand-picked one. Their self-consistency objective is implemented and tested directly here (§3.4).

This paper builds and audits the transport toolkit end to end on the 2D Ising ferromagnet, where a good reaction coordinate is known in advance, and the Edwards–Anderson spin glass, where the obvious candidate carries almost no information, treating each piece of machinery as a hypothesis rather than a technique to be demonstrated. The adjoint=committor identity holds to numerical precision; the empirical claims built on it — that weight windows reach usefully deeper into the tail, that a learned importance coordinate beats a hand-picked one, that FW-CADIS flattens relative error across a family of thresholds, and that any of this earns back its overhead at reachable lattice sizes — fared less uniformly, and are reported as found, with a bootstrap confidence interval attached to every comparative claim.

2. Materials and Methods

Figure 1 summarises the pipeline end to end; the rest of this section describes each stage in turn.

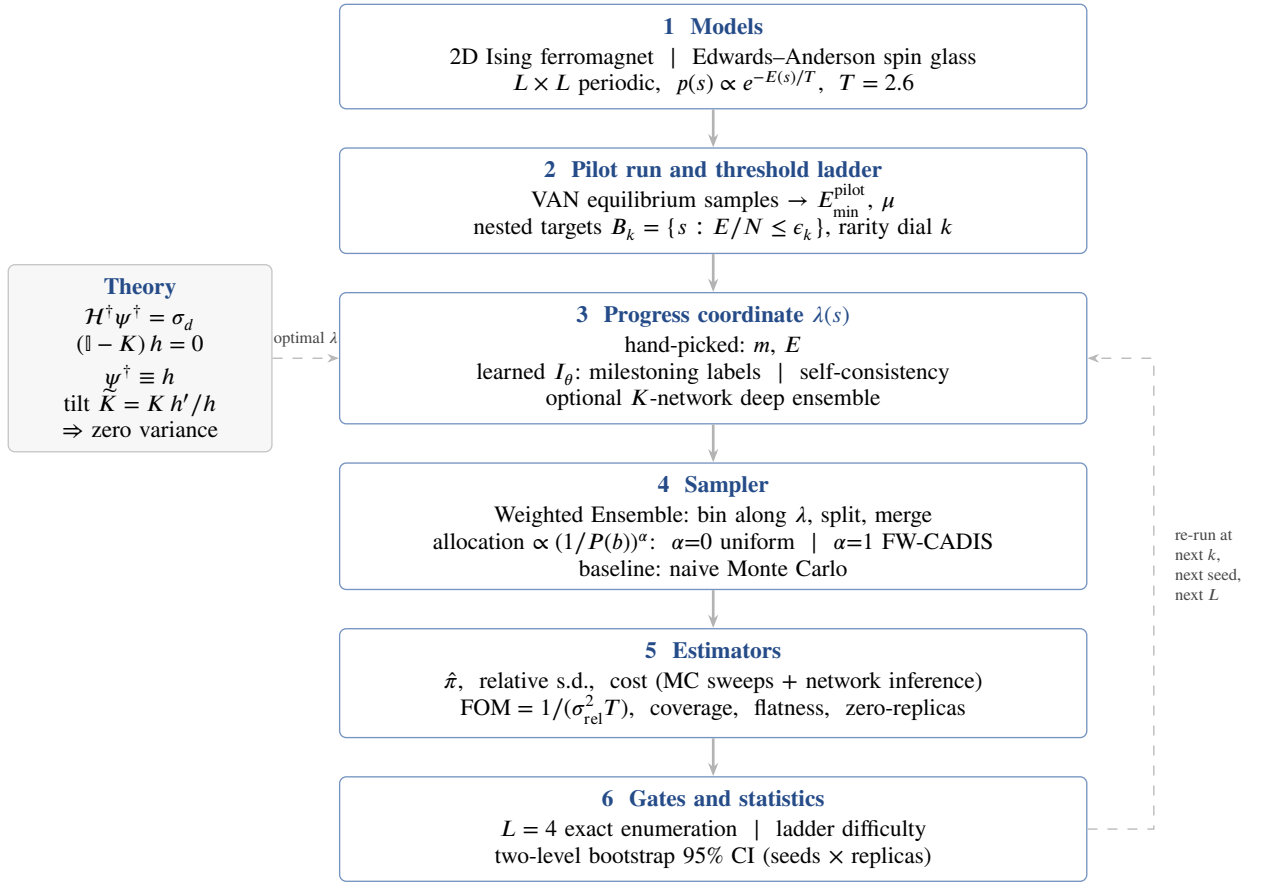


Figure 1: The system end to end. A pilot run fixes a nested ladder of rare-energy targets; a progress coordinate — hand-picked or learned — is chosen; Weighted Ensemble bins walkers along it and splits or merges them under one of two allocation rules; and the resulting estimator is scored on figure of merit, coverage and flatness. Every comparison passes the gates of stage 6 before it is reported. The theory panel (left) is what licenses the whole construction: the transport adjoint and the committor solve the same equation, so tilting by it is zero-variance, and stage 3 is an attempt to approximate that ideal coordinate. The dashed return path is the outer loop over target depth k , network seed and lattice size L .

2.1. Theory: the adjoint=committor identity

2.1.1. Linear transport and the adjoint importance function

Let $P = (\mathbf{r}, \Omega, E)$ be a phase-space point and $\psi(P)$ the particle flux obeying the linear transport (Boltzmann) equation, written abstractly as $\mathcal{H}\psi = q$, with transport operator $\mathcal{H}$ (streaming + collision) and source q . A detector response (“tally”) is a linear functional $R = \langle \sigma_d, \psi \rangle = \int \sigma_d(P) \psi(P) dP$, where σ_d is the detector cross-section. Define the *adjoint* flux $\psi^\dagger$ by

$$\mathcal{H}^\dagger \psi^\dagger = \sigma_d. \quad (1)$$

The defining property of the adjoint gives the fundamental identity

$$R = \langle \sigma_d, \psi \rangle = \langle \psi^\dagger, q \rangle \quad (2)$$

and the physical reading that makes everything work: $\psi^\dagger(P)$ is the expected contribution to the tally R of a single particle introduced at P — its *importance*. If one samples particle transitions biased toward high importance and assigns each particle the statistical weight $w(P) \propto 1/\psi^\dagger(P)$, then, when $\psi^\dagger$ is exact, every history contributes an identical score and the estimator of R has *zero variance*. This is the theoretical ceiling of variance reduction; all practical schemes approximate it.

2.1.2. CADIS, FW-CADIS, weight windows, figure of merit

Since $\psi^\dagger$ is unknown, CADIS (Consistent Adjoint-Driven Importance Sampling) computes a cheap, approximate $\psi^\dagger$ and uses it to drive an expensive forward Monte Carlo run through two consistent devices: **source biasing**, sampling from $\hat{q}(P) = \psi^\dagger(P)q(P)/R$ instead of q ; and **weight windows**, assigning each phase-space cell a target weight $w \propto 1/\psi^\dagger$ and a window $[w_{\text{target}}/k, k w_{\text{target}}]$ — a particle above the window is *split* into equal-weight copies, one below is *Russian-rouletted*. This keeps $w(P)\psi^\dagger(P) \approx \text{const}$, bounding the weight and hence the variance. **FW-CADIS** (forward-weighted CADIS) extends this to *global* variance reduction: weighting the adjoint source by the inverse of the local forward response produces weight windows that equalise *relative* statistical error across many detectors, or the whole domain, at once, rather than optimising a single tally. Efficiency is measured by the figure of merit $\text{FOM} = 1/(\sigma_{\text{rel}}^2 T)$, the inverse product of relative variance and CPU time.

2.1.3. The central identity: importance = committor = zero-variance measure

Work with a Markov chain on configurations s with transition kernel $K(s \rightarrow s')$. Let A (failure/bulk) and B (the rare target set) be disjoint, and define the **committor** $h(s) = \Pr[\text{reach } B \text{ before } A \mid \text{start at } s]$, with $h|_A = 0$, $h|_B = 1$. On the transient set it satisfies the discrete backward equation

$$h(s) = \sum_{s'} K(s \rightarrow s') h(s') \quad \Longleftrightarrow \quad (\mathbb{I} - K)h = 0. \quad (3)$$

Equation (3) is precisely the discrete adjoint transport equation (1) with the target set playing the role of the detector. Hence the rare-event committor $h(s)$, the transport adjoint importance $\psi^\dagger(s)$ (with $\sigma_d = \mathbf{1}_B$), and the “neural importance function” of recent ML papers are the same function: the expected contribution of a walker at s to the rare tally. Defining the tilted kernel $\tilde{K}(s \rightarrow s') = K(s \rightarrow s') h(s')/h(s)$, the path likelihood ratio telescopes to $W = h(s_0)/h(s_{\text{end}}) = h(s_0)$ for every path ending in B : every sample returns the identical value $h(s_0) = \pi$, so the estimator of π has zero variance.

Of the two readings of π available here, this paper always means the second. The *static* equilibrium mass $\Pr_{\text{eq}}(s \in B) = \sum_{s \in B} p(s)$ is a property of the stationary measure alone, answerable by direct enumeration with no dynamics at all — our $L = 4$ gate (§3.1) computes exactly this. The *dynamical* hitting probability $h(s_0)$ from a fixed bulk reference state $s_0 \in A$ is what every $\hat{\pi}$ reported in §3 estimates, and what Weighted Ensemble is built to sample efficiently. The two coincide under the same mixing condition transition-path theory already requires [3]: dynamics equilibrates within A faster than it escapes toward B , which holds throughout the $T = 2.6 > T_c$ regime tested here, though not necessarily at low temperature where A could itself fragment into sub-basins.

The construction was checked numerically on a solvable 1D walk before any spin-system code was written (§3.1); the outcome of that check is why every learned-coordinate result below uses Weighted Ensemble rather than pure importance reweighting.

2.1.4. Scope and limits of the correspondence

The identity above is exact at the level of linear algebra, not merely structurally similar. Production transport codes do not solve the continuous operator equation (1) directly: phase space is first discretized into energy groups and a finite set of directions [8, 9], reducing it to a finite linear system $(\mathbb{I} - \mathbf{K})\psi^\dagger = \sigma_d$ for a transition operator $\mathbf{K}$ packaging streaming and scattering between cells — term for term the same system as Eq. (3). Nothing about K 's physical origin enters that algebra, which is why the identity transfers to spin dynamics with no modification to the mathematics: it holds for *any* Markov process with an absorbing target set. What does not transfer is the physical mechanism generating K . Transport's K encodes free flight interrupted by scattering and absorption at rates set by cross-sections; spin dynamics' K encodes single-spin-flip acceptance under a thermal bath at temperature T . There is no mean free path or cross-section in the spin problem, and a “walker” is a bookkeeping device for one member of a population of independent simulation replicas, not a physical particle.

Two asymmetries follow, in opposite directions. Spin systems are the *easier* problem in one specific sense: multigroup/discrete-ordinates transport carries a discretization error that must be checked against measurement, since the exact continuous answer is rarely available at all, whereas a spin lattice's state space is already finite (2^N configurations, no truncation) — which is why $L = 4$ can be checked against exact enumeration directly (§3.1). They are *harder* in the place that matters most: transport bins its importance function along a natural, low-dimensional, physically meaningful coordinate — position, direction, energy, fixed by the geometry of the shield — while spin

configurations have no such coordinate, and the obvious candidate, magnetization, fails outright on the spin glass (§2.2). That absence, not the linear algebra, is why this paper needs a *learned* importance coordinate (§2.4) at all.

2.2. Models and threshold ladder

Two spin systems are used, both on an $L \times L$ square lattice with periodic boundaries and Gibbs measure $p(s) \propto e^{-E(s)/T}$: the **ferromagnet** ($E = -J \sum_{\langle ij \rangle} s_i s_j$, $J > 0$), where the magnetization m is an obviously informative reaction coordinate; and the **Edwards–Anderson (EA) spin glass** ($J_{ij} = \pm J$ quenched-random per bond), whose low-energy states are disordered and frustrated so that $m \approx 0$ regardless of energy — the case a hand-picked coordinate is expected to fail on, and the one this paper focuses its harder tests on. All results use $T = 2.6$ (above $T_c \approx 2.269$) unless noted; $L = 16$ for full-scale results, $L = 4$ for exact-enumeration gates. The **resampling interval** τ (sweeps between WE split/merge operations) is stated explicitly throughout because it must be comparable to the system’s own correlation time, $\tau_{\text{corr}} \approx 1$ sweep at $T = 2.6$; otherwise splitting is undone by relaxation before the next resampling and every WE[·] variant degenerates silently toward naive Monte Carlo. Results using the full preset (§3.2, §3.4, §3.6) use $\tau = 4$; milestone results (§3.3) use $\tau = 2$.

Threshold ladder. Every rare-event target below is drawn from one nested family rather than chosen ad hoc per experiment. Given a pilot run’s population, let $E_{\min}^{\text{pilot}}$ be its deepest observed energy per spin and μ its bulk mean; the depth- k target set is

$$B_k = \{s : E(s)/N \leq \epsilon_k\}, \quad \epsilon_k = E_{\min}^{\text{pilot}} - k\Delta, \quad \Delta = \frac{\mu - E_{\min}^{\text{pilot}}}{10}, \quad (4)$$

i.e. k steps beyond the pilot’s own most extreme observation, in units of one tenth of its observed dynamic range. This makes k (–beyond throughout) a rarity dial portable across L , T , and disorder draw, and makes B_k nested, $B_1 \supset B_2 \supset \dots$ — the structure both the FW-CADIS threshold family (§3.6) and milestone’s progress labels (§3.3) build on. The dial is genuine in practice, not just in principle: at fixed $L = 16$, $T = 2.6$, $\hat{\pi}$ drops roughly one order of magnitude per step of k , and too large a k places B_k outside what any method can resolve at the compute available, degrading every comparison to noise versus noise.

2.3. Samplers: variational autoregressive network and Weighted Ensemble

Variational autoregressive network (VAN). A normalized autoregressive neural network over spin configurations is trained variationally against the Gibbs measure, so that it draws statistically independent samples by construction rather than by decorrelating a Markov chain. Its role in this project is to remove critical slowing down near T_c , not to sample rare events; the two are different difficulties, and the VAN is used only as an equilibrium sampler.

Weighted Ensemble (WE). WE is the statistical-mechanics name for split/Russian-roulette weight windows: a population of walkers, each carrying a weight, is periodically binned along a chosen coordinate; bins are equalised to a target occupancy by splitting over-weight bins and merging under-weight ones, conserving total weight exactly (hence unbiased for *any* binning coordinate, good or bad). We validate the implementation against exact enumeration at $L = 4$ before trusting any figure-of-merit comparison at scale.

2.4. Learned importance coordinates: two training objectives

A network $I_\theta(s)$ is trained to approximate the committor and then used as the WE binning coordinate, in place of a hand-picked one. Two training objectives are compared.

Surrogate rollout label (this project’s default). Binary committor labels collapse to all-zero for a genuinely rare target, so the network is instead trained to regress the *best progress value reached* in K short rollouts from each collected configuration — a dense, monotone surrogate for the committor, not the committor itself. In its **milestoning** refinement, intermediate thresholds are placed along the ladder from bulk to target and a configuration’s label estimates its committor to the *next* milestone rather than to the final target directly, which stays closer to the theoretical object of §2.1.3 while still avoiding all-zero label collapse.

Self-consistency (Kim & Cai, 2026). A label-free alternative: train I_θ by minimizing violation of its own self-consistency condition under the elementary single-spin-flip kernel, $\sum_j K(s \rightarrow j) I(j) = I(s)$, with boundary condition $I = 1$ on the target. This equation, plus the target boundary alone, is under-determined: the constant solution $I \equiv 1$ satisfies it everywhere, for any amount of training data. A second boundary condition — anchoring bulk-equilibrium states to a distinct reference value — breaks this degeneracy and is necessary, not optional.

2.5. FW-CADIS global allocation

Given a family of K nested thresholds and a cheap pilot estimate of the equilibrium bin occupancy $P(b)$, FW-CADIS allocates walkers per bin $\propto (1/P(b))^\alpha$; $\alpha = 0$ recovers uniform allocation (the standard WE baseline), $\alpha = 1$ is full FW-CADIS. Both settings use identical code, differing only in the allocation vector, which is what makes the comparison fair by construction. The metric is **coverage** first (how many of the K thresholds were observed at all — a method that fails a deep threshold has infinite error there, and scoring only the thresholds it survived would reward failure by making it look artificially flat) and **flatness** second, the spread (max/min) of relative error across the observed thresholds.

2.6. Statistical methodology

Two disciplines are applied uniformly and are largely responsible for this paper’s qualified conclusions rather than overclaimed ones. **Ladder difficulty**: comparing methods in a regime where the baseline itself fails to observe the rare event (many zero-replicas) compares noise to noise, so every comparison below is reported at a depth where the reference method(s) reach 0, or close to 0, zero-replicas out of the replica count, found by backing off from a harder target until this holds. **Bootstrap confidence intervals**: a point-estimate ratio of figures of merit from a handful of replicas is not by itself evidence of a real effect, since FOM is a ratio involving a *sample variance* that is itself noisy at small replica counts. Every comparative claim in §3 is therefore backed by a bootstrap 95% confidence interval, resampling the replica axis with replacement, and claims that do not survive this check are reported as unresolved rather than smoothed into a verdict.

3. Results and Discussion

3.1. Validation against exact references

Nothing below is compared on figure of merit until the machinery producing it has been checked against a reference whose answer is known independently.

The committor identity. The committor identity of §2.1.3 was checked on a solvable 1D absorbing random walk on $\{0, \dots, 15\}$ before any lattice code was written, with the rare event (reaching state 15 before 0) calibrated to $\pi = 1.14 \times 10^{-3}$. Tilting by the exact committor, available there in closed form, gave a measured relative variance of $\sim 10^{-30}$ — zero to machine precision, as the theory requires. Tilting by an *approximate* committor under pure biasing, with no weight windows, was instead worse than naive Monte Carlo. That second outcome is why every subsequent learned-coordinate result in this paper uses Weighted Ensemble (split/roulette) rather than raw importance reweighting: an imperfect importance estimate, used to reweight without bounding the weights, can be actively harmful. It is also why deep-penetration transport uses weight windows rather than relying on reweighting alone.

Weighted Ensemble. At $L = 4$, WE reproduces the exact enumerated $P(m)$ to a mean $|\Delta \ln P|$ of 0.006–0.024 across three tested regimes. At $L = 20$, WE reaches roughly two decades deeper into the magnetization tail than naive Monte Carlo at matched cost: naive’s hard floor is $|m| = 0.886$ ($P \approx 6 \times 10^{-6}$), where it returns zero samples and hence zero information; WE reaches $|m| = 0.932$ ($P \approx 2.4 \times 10^{-7}$). Separately, applying weight windows to that same approximate, otherwise-fragile tilt recovered roughly a $2\times$ figure-of-merit improvement over the same tilt without them (measured FOM $1.24 \times 10^{-5} \rightarrow 4.86 \times 10^{-4}$) — the robustness layer converts an actively harmful bias into a net improvement, exactly as the transport literature predicts. Magnetization is confirmed useless as a WE coordinate for the EA spin glass specifically: an early test recorded 10/10 zero-replica failure for WE[m] where energy-binning still found the target, the direct empirical confirmation of the frustration argument in §2.2.

The VAN. Independently of the rare-event question, the VAN was validated as an equilibrium sampler: trained at $L = 16$, it reproduces the exact Onsager free energy across the tested temperature range. A separate autocorrelation measurement at $L = 32$ found that the integrated autocorrelation time of ordinary Metropolis sampling spikes to $\tau_{\text{int}} = 46.1$ sweeps at $T_c \approx 2.269$, versus 1.5–3 sweeps away from criticality, while the VAN samples independently by construction and pays none of this cost. This is a speed result, orthogonal to the rare-event problem the rest of the paper addresses.

3.2. Learned versus hand-picked coordinates

Figure 2 reports the full-scale ($L = 16$, 10 replicas) result on both models, at a target calibrated one step beyond the pilot’s observed extreme.

Table 1Incremental- R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE 1.0 $\rightarrow$ 0.271 over 20 epochs).

Quantity	value
Pearson $\text{corr}(I_\theta, E/N)$	-0.614
fraction of $\text{Var}(I_\theta)$ unexplained by E/N	0.623
$R^2(\text{label} \sim E/N \text{ alone})$	0.358
$R^2(\text{label} \sim I_\theta \text{ alone})$	0.577
$R^2(\text{label} \sim E/N + I_\theta)$	0.608
incremental R^2 from I_θ beyond E/N	+0.250

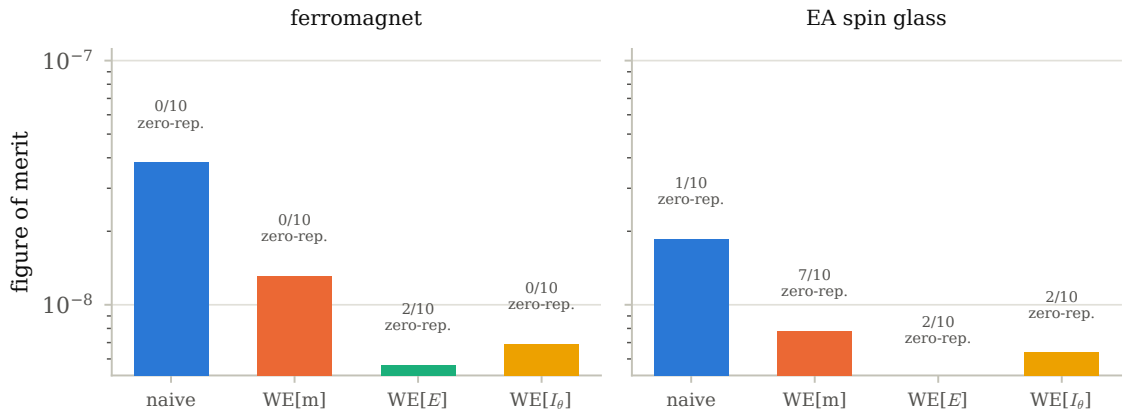
Full scale ($L = 16$, 10 replicas), target one step beyond the pilot's observed extreme

Figure 2: Full scale, both models: FOM by method (log scale) with zero-replica counts annotated, at target $|m| \geq 0.9297$ (ferromagnet) and $E/N \leq -1.0625$ (EA spin glass), 10 replicas. Bar height alone does not tell the whole story — WE[E] has the fewest zero-replicas of any weight-window method on the ferromagnet but a middling FOM, which is why both numbers are reported together.

Two readings are needed. Naive Monte Carlo wins on raw FOM in both models at this rarity ($\sim 4 \times 10^{-5}$); and on the spin glass, with WE[E] and WE[I_θ] on equal footing (both 2/10 zero-replicas), **the learned coordinate has the higher FOM by $\sim 27\%$** (6.36×10^{-9} vs. 5.00×10^{-9}) — the first apples-to-apples advantage for the learned coordinate over the best hand-picked one, and the direct predecessor of the milestone result below. WE[m] fails in most spin-glass replicas (7/10), confirming that magnetization carries essentially no information about this landscape.

That the learned coordinate is not merely a noisy copy of energy is confirmed by a separate diagnostic, which regresses the rare-event label on energy alone, on I_θ alone, and on both, at matched training budget (Table 1). The incremental R^2 of +0.250 is far above the diagnostic's own noise threshold (0.02): I_θ carries substantial genuine information about the spin glass that energy alone does not have, consistent with the FOM advantage measured directly in Fig. 2.

Simply pushing the target rarer does not settle the naive-versus-learned comparison. Rerunning one threshold step deeper leaves 4–10 of 10 replicas failing across every method, including naive itself (4/10, versus 1/10 at the shallower target), so the resulting numbers are noise over noise rather than a result. This run established the ladder-difficulty discipline of §2 for the remainder of the project, and motivated the milestone objective below — a change to the training label rather than to the weight-window schedule.

Table 2

Reference methods, EA spin glass, $L = 16$, $T = 2.6$, one step beyond the pilot extreme, 10 replicas, milestoning labels. All three are bit-identical across every training run, since none depends on the trained network. The corresponding $\text{WE}[I_\theta]$ results — eighteen independent runs of the identical configuration, seventeen with per-replica data retained — are shown individually in Fig. 3; seed 0 was run twice and reproduced bit-for-bit.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	1.48×10^{-7}	0/10
$\text{WE}[m]$	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
$\text{WE}[E]$	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10

3.3. Milestoning labels: a clean regime, but strong seed dependence

On the EA spin glass at $L = 16$, one threshold step beyond the pilot’s observed extreme (“beyond=1”), every method reaches **0/10 zero-replicas** — the cleanest, most-trustworthy regime obtained in this project by the ladder-difficulty standard (Table 2).

Every $\text{WE}[I_\theta]$ run is, as far as the pipeline is concerned, the *same experiment*: identical target, identical training data from cached labels, identical hyperparameters. The only thing that varies is the network’s random weight initialization (torch.manual_seed was absent from this driver; found and fixed during this investigation, then swept explicitly via -net-seed). **Seventeen seeds give seventeen different outcomes, ranging from a 52% loss to a 112% win against $\text{WE}[E]$** — a wide, roughly symmetric spread around a small positive average. The seeding fix is complete: seed 0 was run twice and reproduced bit-for-bit, including the full 1774–1776 s wall-clock trajectory, ruling out any other unseeded source of variation.

Pooling the seventeen with a two-level bootstrap — outer resample over seeds, inner resample over each seed’s 10 replicas — gives a geometric-mean FOM ratio of $1.19\times$ with 95% CI [0.84, 1.75] (median-of-ratios agrees: $1.20\times$, CI [0.74, 1.85]; the arithmetic mean of ratios is unstable at $R = 10$, because bootstrap draws landing on near-duplicate resamples give rel. s.d. near zero and, since $\text{FOM} \propto 1/\text{rel. var.}$, produce large outliers). Tracked at four checkpoints as seeds were added — $1.28\times$ ($n = 7$), $1.17\times$ ($n = 9$), $1.28\times$ ($n = 12$), $1.19\times$ ($n = 17$) — the estimate does *not* trend in either direction; what moves monotonically is the CI width ($1.73 \rightarrow 1.36 \rightarrow 1.18 \rightarrow 0.91$), the expected signature of averaging down noise around a small effect whose sign is not resolved by seed count alone. Figure 3 shows both the per-seed spread and the checkpoint trajectory. Naive Monte Carlo has the best raw FOM in every run at this rarity ($\sim 5 \times 10^{-5}$) regardless of seed, the same lesson demonstrated in miniature by the 1D toy walk of §3.1.

Milestoning: $\text{WE}[I_\theta]$ vs. $\text{WE}[E]$, $L = 16$, beyond = 1

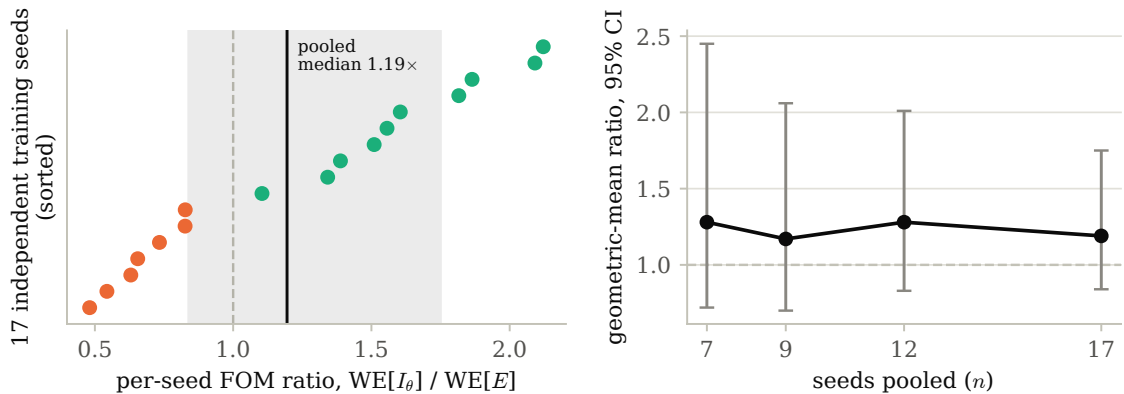


Figure 3: Left: all 17 per-seed FOM ratios (sorted), with the pooled two-level-bootstrap median and 95% CI shaded. Right: the four tracked checkpoints as seeds were added — the point estimate bounces while the CI width shrinks monotonically, the expected signature of averaging down noise around a small, sign-unresolved effect.

Table 3

 EA spin glass, $L = 16$, one step rarer than Table 2, milestone labels, default configuration ($n_{\text{bins}} = 50$, $R = 10$).

Method	FOM	zero-rep.
naive	3.51×10^{-8}	0/10
WE[m]	7.73×10^{-9}	0/10
WE[E]	9.21×10^{-9}	0/10
WE[I_θ]	8.36×10^{-9}	0/10

Table 4

 Five independent $K=5$ deep-ensemble runs, identical configuration to Table 2. FOM ratio is $\text{WE}[I_\theta]/\text{WE}[E]$ on the MC-sweep cost as originally defined.

Run	WE[I_θ] FOM	rel. s.d.	ratio vs. WE[E]
ensemble, base seed 20	1.95×10^{-8}	0.272	$0.46\times (-54\%)$
ensemble, base seed 21	2.75×10^{-8}	0.229	$0.65\times (-35\%)$
ensemble, base seed 22	3.69×10^{-8}	0.199	$0.88\times (-12\%)$
ensemble, base seed 23	3.07×10^{-8}	0.215	$0.73\times (-27\%)$
ensemble, base seed 24	3.84×10^{-8}	0.194	$0.91\times (-9\%)$

A plausible infrastructure explanation for the spread was tested and ruled out. One candidate cause is that $\text{WE}[I_\theta]$'s bin range was left adaptive (auto-ranging from the population actually explored) rather than calibrated to the coordinate's own scale, unlike $\text{WE}[E]$ and $\text{WE}[m]$. Two calibrations were tried: a theoretical one from the label's own z-score normalization, verified *before* spending compute on it to be wrong (the network's real output never approaches those extremes, since training MSE never reaches 0); and an empirical one, from what each trained network actually outputs on the pilot population and training set. Run head-to-head against the untouched adaptive baseline on matched net-seeds (identical trained network per pair, only the bin range differing), the empirical calibration made both the mean FOM ratio ($0.79\times \rightarrow 0.61\times$) and the rel. s.d. ($0.214 \rightarrow 0.286$) *worse* across seeds 1–3. A fixed range clips any walker that exceeds it into the boundary bin, while adaptive tracking never does; for a coordinate whose true dynamic range over a multi-thousand-iteration WE trajectory is not well estimated from a static pre-run snapshot, that clipping costs more resolution than the calibration gains. The seed variance is therefore intrinsic to what each trained network learns, not an artifact of how its output is binned.

Bin-count tuning is likewise ruled out as a fix. One threshold step rarer (Table 3), $\text{WE}[I_\theta]$ scores worse than $\text{WE}[E]$ in every configuration tried: at the default bin count, at double the replica count ($R = 10 \rightarrow 20$, same $n_{\text{bins}} = 50$, same qualitative pattern), and at a $1.8\times$ larger bin count ($n_{\text{bins}} = 50 \rightarrow 90$), which made $\text{WE}[I_\theta]$'s *own* FOM worse rather than better ($8.36 \times 10^{-9} \rightarrow 5.36 \times 10^{-9}$). A genuine resolution shortfall would improve with more bins, not degrade. The mechanism is structural: the learned coordinate's walker-population cost scales with bin count, while a discretized coordinate like energy naturally saturates a small, fixed number of bins and does not pay that overhead as it grows.

3.3.1. Deep ensembling: the seed variance is reducible, but converges to a loss

Table 2 leaves one question to settle: is the seed-to-seed spread genuine noise around a real advantage, or is the positive point estimate itself an artifact of a few lucky initializations? Rather than add more single-net seeds — already shown not to close the CI at reasonable cost — we attacked the diagnosed noise source directly. `run_milestone.py` was extended with `-ensemble K`: train K independently-initialized networks on the identical milestone labels and use their *mean* prediction as I_θ , a standard deep-ensemble variance-reduction step. $K = 1$ reproduces the original single-net code path exactly (verified bit-identical output on a smoke config). Five independent ensembles ($K = 5$ each, base seeds 20–24) were run at the identical configuration as Table 2 (beyond=1, $\tau = 2$, $R = 10$).

Two findings follow, both against the ensemble. **First**, ensembling did what it was supposed to do to the *variance*: the five ratios of Table 4 span $0.46\text{--}0.91\times$, far tighter than the seventeen single-net seeds' -52% to $+112\%$ spread. But the value it converges toward is a loss, not the earlier $\sim 1.2\times$ point estimate: a two-level bootstrap (outer over the 5 runs, inner over each run's 10 replicas, identical machinery to Table 2) gives a geometric-mean ratio of $0.74\times$, 95% CI $[0.43, 1.32]$ — not significant, but on the opposite side of 1 from the single-net pooled estimate. Read together,

Table 5

EA spin glass, $L = 16$, shallowest calibrated target, 10 replicas. Point estimates only — see the bootstrapped intervals in Fig. 4.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	3.36×10^{-5}	0.294	8.60×10^{-8}	0/10
WE[m]	3.94×10^{-5}	0.553	8.81×10^{-8}	1/10
WE[E]	3.39×10^{-5}	0.671	1.33×10^{-8}	0/10
WE[I_θ] surrogate	2.92×10^{-5}	0.904	8.62×10^{-9}	1/10
WE[I_θ] self-consistent	4.01×10^{-5}	0.655	1.62×10^{-8}	0/10

the most consistent interpretation is that the single-net sweep’s more extreme positive draws (+82%, +109%, +112%) were high-variance artifacts of individual weight initializations rather than signal, and I_θ ’s true expected performance is at or below WE[E]’s.

Second, and decisive, the ensemble exposed a cost-accounting bug that applies to every WE[I_θ] result in this paper, though it is negligible at $K = 1$. `we_estimate()`’s `cost` field counts Monte Carlo sweeps only ($|S| \cdot \tau \cdot L^2$) and is blind to neural-network inference. At $K = 1$ a single forward pass is cheap enough relative to the MC cost that the omission does not matter; at $K = 5$ it does. Measured directly from wall-clock timestamps rather than the MC-cost field, WE[I_θ] took 10.7–10.9× WE[E]’s wall-clock time in every one of the five runs — remarkably consistent — while the MC-cost field implied only $\sim 1.3\times$. Recomputing FOM with measured wall-clock cost, using the same bootstrap, gives a geometric-mean ratio of 0.087×, 95% CI [0.051, 0.155]: the only comparison in this paper whose confidence interval *excludes* 1.0 in either direction. `run_milestone.py` now records measured `wall_time` per method, so the correction no longer requires post-hoc log parsing.

Deep ensembling therefore does not rescue the learned coordinate. It reduces run-to-run variance exactly as a deep ensemble should, but converges to a value at or below parity with the hand-picked energy coordinate, and once its real inference cost is counted at all the comparison is a decisive, statistically significant loss.

3.4. The self-consistency objective of Kim and Cai

The self-consistency training objective of §2.4 was implemented, gated against exact enumeration at $L = 4$ (learned-coordinate unbiasedness ratio 1.004), and run head-to-head against the surrogate objective, first at laptop scale and then at full scale. At $L = 12$ and the shallowest calibrated target, the self-consistency objective’s point estimate already exceeded the surrogate’s (4.98×10^{-8} vs. 2.93×10^{-8} , with fewer zero-replicas, 2/6 vs. 3/6), though it still fell short of WE[E]’s 7.53×10^{-8} at this rarity. Its training cost is real even here: $\sim 94\times$ longer than the surrogate objective at this scale (~ 752 s vs. ~ 8 s), from the additional $N = L^2$ network evaluations the self-consistency loss requires per training configuration.

At full scale ($L = 16$) and the shallowest calibrated target — “beyond” = 0, the easiest regime, in which every method reaches 0–1 zero-replicas out of 10 — the self-consistency coordinate’s *point estimate* exceeds both the hand-picked energy coordinate and this project’s own surrogate coordinate (Table 5).

That point estimate does not survive a bootstrap check, and the effect shrinks under replication. Both configurations were re-run with per-replica values retained, and the FOM ratio between methods bootstrapped by resampling the replica axis independently for each side — no shared random-number seed to pair on, since uniform-style and biased allocations consume different numbers of walkers per bin and desynchronise almost immediately, the same obstruction encountered independently in the FW-CADIS common-random-numbers attempt of §3.6. Because training here is a fixed cost independent of replica count (a deterministic seed; only the cheap evaluation step scales with R), tripling to $R = 30$ cost roughly six additional minutes on top of the ~ 76 -minute training already paid. Figure 4 reports the result, together with the same comparison run one threshold step deeper (depth 1, the depth of Table 2), where the zero-replica counts are 2/10 for the surrogate against 1/10 for the self-consistent objective.

No ratio is significant at any depth or replica count. At $R = 30$ both confidence intervals tighten substantially and both median ratios move toward 1 — against the surrogate from $1.83\times$ ([0.49, 9.23]) to $1.39\times$ ([0.65, 2.95]), and against WE[E] from $1.19\times$ ([0.33, 4.95]) to $1.07\times$ ([0.55, 2.09]) — not merely a tightening interval around the original estimate, but the point estimate itself receding, which we read as evidence that the original $R = 10$ advantage was substantially noise. The deeper target tells the same story from the other direction: the point-estimate gap versus

the surrogate *shrank* ($1.83\times \rightarrow 1.14\times$) rather than growing, even as the self-consistency coordinate kept its relative reliability advantage (fewest zero-replicas of any weight-window method at both depths). One more noisy $R = 10$ point estimate does not resolve the depth question either way, but it does rule out the simplest optimistic story, that the advantage would obviously grow once naive and the surrogate objective start to struggle.

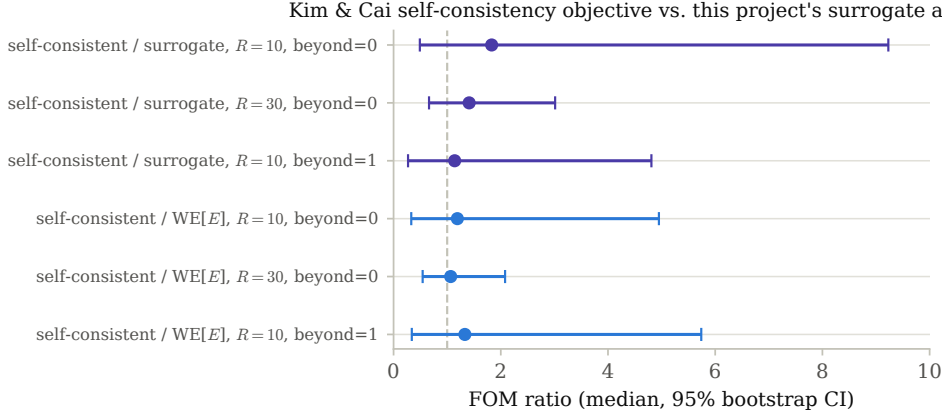


Figure 4: Bootstrapped FOM ratios, full scale ($L = 16$), as a forest plot: the self-consistency objective against the surrogate objective and against WE[E], at depth 0 ($R = 10$ and $R = 30$) and depth 1 ($R = 10$). A ratio above 1 favors the self-consistency objective. Every interval crosses the ratio-1 reference line (dashed); the $R=30$ intervals (recomputed directly from the retained per-replica arrays via the same replica-resampling bootstrap) are visibly tighter than their $R=10$ counterparts without the median crossing to the other side of 1.

In summary, the self-consistency-trained coordinate is the most reliable weight-window method tested in every regime examined, and its point estimate is consistently the best of the learned-coordinate variants, but a quantified FOM advantage over this project's own surrogate objective is not established at any depth or replica count tested and shows no sign of strengthening with depth. It is also markedly more expensive to train: roughly $170\times$ slower per epoch than the surrogate objective at full scale (~ 76 minutes versus ~ 4).

3.5. Lattice-size dependence of naive dominance

Both §3.2 and §3.3 find naive Monte Carlo winning on raw FOM at $L = 16$. This is better read as a finite-size effect than as a failure of variance reduction. Naive's cost is fixed by the replica budget (naive_chains $\times$ naive_sweeps in our runs) independent of the target's rarity π , while its relative variance grows as π shrinks; WE's cost is likewise fixed by its own budget ($n_{\text{bins}} \times n_{\text{per bin}} \times \text{we_iter}$), but once it penetrates to the target its relative variance is largely rarity-independent. The two cost curves must cross somewhere, and at a rarity shallow enough for naive to sample the event comfortably, naive wins almost by construction, since WE pays a penetration overhead the problem does not yet need. We tested this directly with a lattice-size scan at fixed $T = 2.6$, beyond = 1, $\tau = 2$: $L \in \{8, 16, 24, 32\}$, with naive_chains/naive_sweeps held fixed across all four so that naive's cost is genuinely L -independent as the argument requires (Fig. 5).

Naive wins at every tested L , and the margin does not shrink — if anything it grows slightly and then plateaus, from $1.9\times$ at $L = 8$ to $\sim 3.5\times$ by $L = 16$ and beyond. This holds across nearly five orders of magnitude of target rarity ($\hat{\pi}$ from 1.8×10^{-4} down to 4.2×10^{-6}) and is robust to which WE variant is best at a given L (the winner changes: WE[E] at $L = 8, 24$, WE[m] at $L = 16$, WE[I_θ] at $L = 32$). Within the range tested there is no sign of the naive/WE crossover opening as L grows; if it exists at all on this problem, it lies beyond $L = 32$ at this temperature. The precise claim supported here is therefore narrower than “variance reduction does not pay off”: it is that no tested rarity was found, at any L up to 32, at which naive fails to converge while a WE-based method succeeds.

Table 6

EA spin glass, $L = 16$, full preset, $R = 30$, 6 thresholds spanning progress $0.691 \rightarrow 1.063$. Every allocation reaches full 6/6 coverage at this scale.

Allocation	flatness (max/min)	cost
uniform ($\alpha = 0$)	298.0 (best)	1.55×10^8
$\alpha = 0.25$	377.8	7.89×10^7
$\alpha = 0.5$	323.9	2.35×10^7
$\alpha = 0.75$	342.4	1.98×10^7
$\alpha = 1.0$	385.5	1.94×10^7 ($\sim 8\times$ cheaper)
naive (reference)	445.1	1.34×10^8

Naive vs. best weight-window variant across lattice size, beyond $= 1$, $\tau = 2$

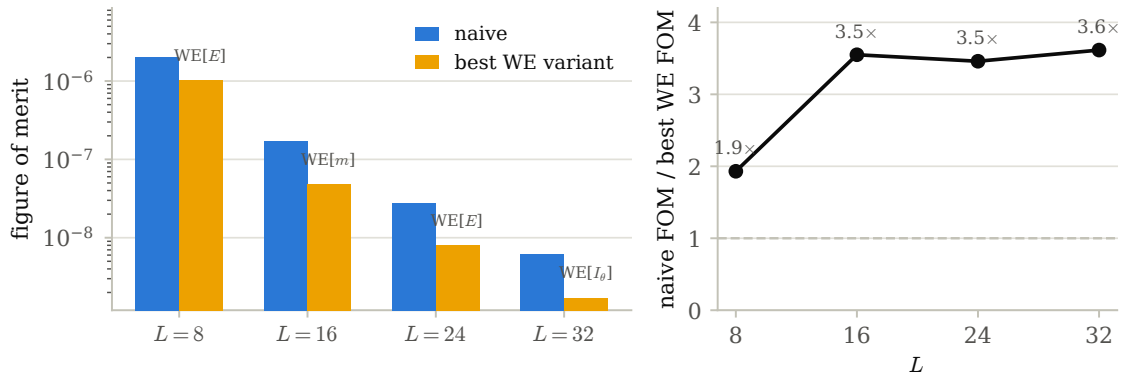


Figure 5: Lattice-size scan, EA spin glass, beyond $= 1$, $T = 2.6$, $\tau = 2$. Left: naive vs. the best-performing WE variant at each L , FOM on a log scale, labeled with which variant won; naive’s $\hat{\pi}$ falls from 1.79×10^{-4} at $L = 8$ to 4.17×10^{-6} at $L = 32$. Right: the ratio itself — it opens from $L = 8$ to $L = 16$ and then plateaus around $3.5\times$, with no sign of narrowing back toward the naive/WE crossover within the tested range.

One secondary pattern is visible but reported only as a point estimate: WE[I_θ] is the weakest WE variant at $L = 8, 16, 24$ but the *strongest* at $L = 32$, consistent with a learned coordinate’s advantage over hand-picked ones mattering more as the landscape gets harder. We do not treat this as established, because the milestone network’s training is not seeded in this scan, so each WE[I_θ] point in Fig. 5 is a single random draw from an uncharacterized distribution over training outcomes. The tested range also has a clear upper edge: at $L = 40$, naive itself fails in 10 of 12 replicas, with WE[m] (11/12) and WE[E] (10/12) failing similarly badly, so by the ladder-difficulty discipline of §2 that size is not a trustworthy comparison as run and is not reported as a fifth point.

3.6. FW-CADIS global allocation: cost advantage without flattening

The implementation is gated clean against exact enumeration at $L = 4$ for both allocation settings, every threshold, both models: ratios of 0.999–0.998–0.989 (uniform) and 1.005–1.012–0.992 (FW-CADIS, $\alpha = 1$) on the ferromagnet, and 1.009–1.004–1.007 / 0.986–0.980–0.983 on the spin glass, all within $\sim 2\%$ of exact — confirming that all thresholds are correctly tallied from a single run regardless of allocation, the core mechanic the whole method depends on.

At laptop scale ($L = 12$, $R = 6$, 6 thresholds) the sweep over α produced a real-looking win: $\alpha = 0.25$ was the only setting with full 6/6 coverage and better flatness than uniform, a modest $\sim 14\%$ improvement ($283.8 \rightarrow 245.2$ in max/min relative error) at $\sim 2.6\times$ lower cost. Larger α looked dramatically flatter but was disqualified by the coverage-first rule of §2, having stopped observing the deepest threshold at all. **The identical sweep at full scale reverses that finding** (Table 6).

At full scale *every* tested α is worse than uniform on flatness, and not even monotonic in α among the FW-CADIS runs ($\alpha = 0.5$ beats both 0.25 and 1.0). All are substantially cheaper ($2\text{--}8\times$), so the method is not doing nothing, but

Table 7

EA spin glass, $L = 16$, shallowest calibrated threshold family, $R = 90$: the final, most favorable configuration tested for FW-CADIS.

Allocation	flatness (max/min)	95% CI	cost
Uniform ($\alpha = 0$)	276.1	[224, 344]	1.59×10^8
FW-CADIS ($\alpha = 1$)	297.3	[234, 388]	1.99×10^7 ($\sim 8\times$ cheaper)

Table 8

Ferromagnet, $L = 16$, shallowest calibrated threshold family, $R = 30$ — the landscape where the coordinate is known-good, unlike the spin glass used everywhere else in this section.

Allocation	flatness (max/min)	95% CI	cost
Uniform ($\alpha = 0$)	100.9	[65.8, 160.3]	1.96×10^8
FW-CADIS ($\alpha = 1$)	287.3	[194.7, 414.8]	3.09×10^7

it is not delivering the flattening it exists to demonstrate at this lattice size, bin count, and threshold spacing. Every remaining lever was then tried against that finding: bootstrap confidence intervals on the flatness metric; a replica increase from $R = 30$ to $R = 90$; wiring the learned coordinate of §3.3 in as the FW-CADIS binning axis instead of the physical progress coordinate (gated clean at $L = 4$, tested at scale: no improvement, roughly double the cost); a common-random-numbers paired-difference design, which does *not* engage on this problem (correlation between the two allocations' same-seed estimates measured ≈ -0.05 to -0.10 , essentially zero, because the two allocations consume different numbers of walkers per bin and desynchronise the random stream almost immediately after the first resampling step); and laddering the threshold family to its shallowest calibrated depth, combined with the larger replica count. The most favorable configuration reached is Table 7.

The two confidence intervals overlap, and the paired-difference estimate is $+20.9$ with 95% CI $[-77.2, +137.6]$, also including zero. Across every α , replica count, and coordinate tried, the point-estimate *direction* consistently favored uniform allocation over FW-CADIS on raw flatness, never the reverse. The confidence-interval width scaled almost exactly as $1/\sqrt{R}$ between $R = 30$ and $R = 90$ (predicted shrink factor 0.577, observed 0.597); extrapolating that scaling, fully resolving the residual gap would need on the order of $R \approx 2300$ replicas, and given the consistent direction such a run would most likely confirm uniform allocation as measurably flatter rather than produce a win for FW-CADIS. **The result we report is therefore that on this problem FW-CADIS does not measurably flatten relative error across a threshold family better than uniform allocation, but achieves a statistically indistinguishable error profile at approximately one-eighth the simulation cost.** The cost advantage is real and reproducible; the flattening advantage the method is named for is the part that did not materialize.

This is not a property of the adversarial spin-glass landscape. Repeating the identical comparison on the ferromagnet, where the reaction coordinate is known to be good and the equilibrium landscape smooth (Table 8), the two confidence intervals do *not* overlap: the paired-difference estimate is $+178.2$ with 95% CI $[+59.8, +351.3]$, excluding zero. **On the easier landscape FW-CADIS is not merely inconclusive but measurably, statistically confirmed worse than uniform allocation** — the opposite of what a landscape-difficulty explanation would predict, since the spin glass, where the method's own hypothesis says a global map should matter most, never reached this level of statistical resolution in either direction. The per-threshold error profiles behind both comparisons are shown in Fig. 6. Common-random-numbers correlation here measured 0.062, consistent with zero and with the mechanism identified above: the two allocations consume different walker populations per bin regardless of which model is sampled. The explanation therefore lies in this specific parameterisation — bin count, threshold spacing, or the inverse-response weighting itself at this scale — rather than in spin-glass frustration.

Per-threshold relative error: uniform vs. FW-CADIS allocation

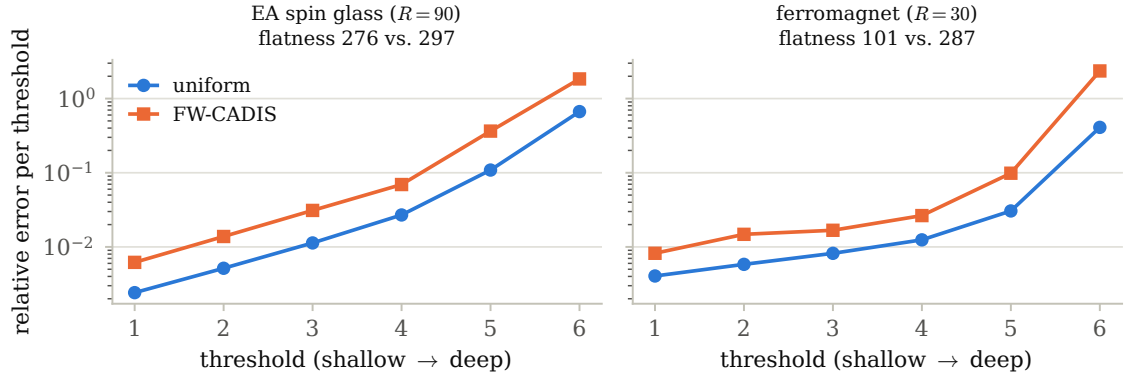


Figure 6: Per-threshold relative error (log scale), uniform vs. FW-CADIS allocation, at the final tested configuration for each model (Table 7 spin glass, Table 8 ferromagnet). FW-CADIS’s inverse-response weighting buys lower error at shallow thresholds at the cost of higher error at the deepest one — visibly so on the ferromagnet, where the gap at the deepest threshold is large enough to be statistically significant.

4. Conclusions

The transport–statistical-mechanics correspondence this project set out to test is real where it can be proved. The adjoint=committor identity is exact, tilting by the exact committor is provably zero-variance and was confirmed numerically to $\sim 10^{-30}$ relative variance on a solvable toy walk, and Weighted Ensemble — weight windows applied to spin configurations — reproduces exact enumeration at $L = 4$ to 1–2% while reaching about two decades deeper into a magnetization tail than naive Monte Carlo at matched cost. The more experimentally demanding claims built on top of that foundation did not hold up as cleanly. The learned coordinate trained on milestone committor labels does not beat the hand-picked energy coordinate: a deep-ensemble follow-up, designed to resolve the single-net seed sweep’s ambiguous $1.19\times$ point estimate by attacking its diagnosed cause, settled it in the negative at $0.087\times$ (95% CI $[0.051, 0.155]$) on a corrected cost accounting, the only bootstrap interval in this paper that excludes parity. FW-CADIS delivers a real, reproducible cost advantage of roughly $8\times$ but not the flattening it is named for, on either landscape. The self-consistency objective is the most reliable weight-window method tested but shows no quantified FOM advantage that survives replication. Naive Monte Carlo’s dominance over every tested weight-window variant, meanwhile, is the one claim that grew *more* confident under additional scrutiny, holding across a four-point lattice-size scan up to $L = 32$ with no sign of narrowing.

The clearest methodological lesson of this paper is not any single number: it is that “0/10 zero-replicas” and a tidy point-estimate table are not, by themselves, evidence of a real effect. The FW-CADIS flattening claim and the Kim & Cai comparison both needed a bootstrap confidence interval to reveal that their apparent wins were noise. The milestone result needed something no amount of replica-axis bootstrapping within a single run could have caught — independently-trained networks — and even the most striking of the seventeen point estimates ($+112\%$) turned out to sit inside a two-level bootstrap interval that includes 1. What finally resolved that question was not more of the same axis but a different one: a deep ensemble that reduced the seed-to-seed spread as a variance-reduction technique should, converged at or below parity, and in the course of being built fairly surfaced a cost-accounting bug whose correction turned a marginal-looking loss into a decisive one. That a genuinely decisive answer came from fixing what was being measured, rather than from sampling harder along an axis already shown not to help, is the most transferable lesson here. Reaching these conclusions required diagnosing and correcting real implementation issues along the way — a resampling interval that silently degenerated a weight-window method to naive sampling, a mis-calibrated bin range, an under-determined training objective that could converge to a useless constant solution while reporting a plausible loss, and an unseeded network initialization that let one training run’s outcome be mistaken for a stable result — each caught by an independent cross-check rather than by a number simply looking wrong.

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